

# Exam Instructions

## Programming and Quantum Computing PGR 213

### Spring 2025

The exam will be oral, and take place on-location for Oslo students and digitally for Bergen students during the week June 10-17. The exam evaluators will be

1) *Gabriel Balaban, Associate Professor in Computer Science, School of Economics Innovation and Technology, Kristiania University of Applied Sciences*

2) *Gunnar Lange, Postdoctor in Physics, Section for Material Physics and Quantum Technology, Department of Physics, University of Oslo*

For the exam you should prepare **one** of the following two cases.

#### **Case 1 : Noisy Quantum Communication**

*You are an engineer working for the start-up company “Teleportation Inc” that specializes in quantum communications. The company CEO would like to set up a qubit teleportation service between Oslo and Trondheim, but is concerned about how random errors might effect the reliability of the service. The CEO would like you to **simulate** a quantum teleportation service, **investigate** the effects of random errors, and **evaluate** potential error correction strategies to help in the design of the service.*

#### **Case 2 : Breaking RSA**

*You are working for the Norwegian Intelligence Service (Etterretningstjeneste) and have been asked by a senior cybersecurity analyst to **demonstrate** how RSA encryption can be broken with a quantum computer and Shor’s algorithm. The analyst is also wondering why RSA encryption cannot already be broken today with Shor’s algorithm? What are the major limitations and or bottlenecks?*

Hint: (Use small values of  $p$  and  $q$  when implementing Shor’s, then investigate the challenges that would be involved for larger values of  $p$  and  $q$ ).

You should implement a set of circuits and algorithms (that you select) in a jupyter notebook in a way that best addresses the case.

#### **Exam Format**

The exam format will be a 3-part oral examination which will take circa 25 minutes to complete.

#### **Part 1 – Presentation of your Jupyter notebook (8 minutes)**

Your notebook should **demonstrate** the main quantum algorithms and circuits that you choose to address your case, along with some experimental calculations involving your chosen algorithms. You should use appropriate visual diagrams and visual data representations to present your results as there will not be enough time to examine individual lines of code in detail.

For the presentation we suggest a three part format, which you can use or modify as you wish:

- a) **Intro** - Introduce the case and give an overview of your solution.
- b) **Evidence and Demonstrations** - Present your chosen algorithms and the results of running them.
- c) **Conclusions** – Describe what can be concluded based on your presented evidence.

### **Part 2 – Whiteboard concept mapping (8 minutes)**

Draw a concept map on the whiteboard demonstrating the major quantum computing concepts involved in your case solution, and their relationships. An example concept map is provided in the appendix. For examples of concept maps involving the course material you can watch the video recordings for lectures 8 and 10-14 in canvas.

### **Part 3 – Whiteboard calculations (8 minutes)**

Perform some calculations on the whiteboard demonstrating key steps of your chosen quantum algorithms. You may also get asked to perform specific calculations by the evaluators.

It is recommend that for each part, you plan on presenting for circa 5 minutes, and leave 3 minutes for questions.

### **Exam Equipment:**

For Part 1 you can use a *Jupyter notebook*. You can save your notebook with the calculations complete so that you do not need to rerun them live during the exam.

For Oslo students there will be a laptop in the room with the notebook. For Bergen students the exam department will set up a special room in Bergen for a remote exam.

You should submit your notebook via the canvas assignment “[Exam jupyter notebook submission](#)” before the exam so that the notebooks can be prepared for you in advance. This will allow us to deal with any potential technical issues in advance of the exam day.

For parts 2-3 only a whiteboard, markers and your knowledge are needed.

### **Tips:**

- 1) Focus on *demonstrating* your knowledge and skills, in a way that addresses the case.
- 2) Practice your presentation in advance to perfect your timing. Time will pass quickly during the examination.

Appendix

Example concept map based on Star Wars.



